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Original Article

Lasting Effects of Cancer and Its Treatment on Employment and Finances in Adolescent and Young Adult Cancer Survivors

Tyler G. Ketterl, MD, MS [1,2]; Karen L. Syrjala, PhD[1]; Jacqueline Casillas, MD, MSHS[3]; Linda A. Jacobs, PhD, CRNP[4]; Steven C. Palmer, PhD[4]; Mary S. McCabe, RN, MA[5]; Patricia A. Ganz, MD[3]; Linda Overholser, MD[6]; Ann Partridge, MD, MPH [7]; Emily Jo Rajotte, MPH[1]; Abby R. Rosenberg, MD, MS, MA [2]; Betsy Risendal, PhD[6]; Donald L. Rosenstein, MD[8]; and Kevin Scott Baker, MD, MS[1,2]

BACKGROUND: The impact of cancer and its treatment on employment and financial burden in adolescents/young adults (AYAs) is not fully known. **METHODS:** Eligibility for this cross-sectional study of AYA cancer survivors included the diagnosis of a malignancy between ages 18 and 39 years and survey completion within 1 to 5 years from diagnosis and ≥1 year after therapy completion. Participants were selected randomly from the tumor registries of 7 participating sites and completed an online patient-reported outcomes survey to assess employment and financial concerns. Treatment data were abstracted from medical records. Data were analyzed across diagnoses and by tumor site using logistic regression and Wald-based 95% confidence intervals adjusting for age (categorized), sex, insurance status, education (categorized), and treatment exposures. **RESULTS:** Participants included 872 survivors (breast cancer, n = 241; thyroid cancer, n = 126; leukemia/lymphoma, n = 163; other malignancies, n = 342). Exposure to chemotherapy in breast cancer survivors was associated with an increase in self-reported mental impairment in work tasks (odds ratio [OR], 2.66) and taking unpaid time off (OR, 2.62); survivors of "other" malignancies reported an increase in mental impairment of work tasks (OR, 3.67) and borrowing >$10,000 (OR, 3.43). Radiation exposure was associated with an increase of mental impairment in work tasks (OR, 2.05) in breast cancer survivors, taking extended paid time off work in thyroid cancer survivors (OR, 5.05), and physical impairment in work tasks in survivors of "other" malignancies (OR, 3.11). Finally, in survivors of "other" malignancies, having undergone surgery was associated with an increase in physical (OR, 3.11) and mental impairment (OR, 2.31) of work tasks. **CONCLUSIONS:** Cancer treatment has a significant impact on AYA survivors' physical and mental work capacity and time off from work. ***Cancer* 2019;125:1908-1917**. *© 2019 American Cancer Society.*

KEYWORDS: adolescent/young adult, cancer, employment, finances, survivorship.

INTRODUCTION

The incidence of cancer in adolescents and young adults (AYAs) aged 15 to 39 years has steadily increased over the past 25 years, and cancer remains a leading cause of nonaccidental death in this age group.[1] Despite advances in cancer prevention, early detection, and treatment over the past several decades, survival rates for AYAs have not improved to the extent they have for younger children or older adult cancer populations.[2,3] AYA survival is often accompanied by ongoing physical, mental, social, and emotional challenges, such as physical impairment, infertility, uncertainty, fear of recurrence, interruption of life plans, and discrimination in employment and insurance.[4,5]

AYAs with a history of cancer were more likely not to be working because of illness compared with controls.[6] AYA survivors who had pursued careers involving physical abilities before a cancer diagnosis often believe that they need to adjust their goals as a result of their cancer.[7] In addition, AYA cancer survivors have higher direct annual medical costs ($7417 vs $4247 for adults without a cancer history), experience annual average lost productivity costs because of illness/disability of $2200 per year, and have lower family incomes compared with similar-age adults without cancer.[6] These disparate outcomes in the AYA population are likely related to the distinctive disruptions and challenges related to education, employment, physical limitations, identity, relationships, and family experienced by the AYA survivor population.

The LIVESTRONG (LS) Foundation developed the Survey for People Affected by Cancer to ascertain how cancer has affected the lives of cancer survivors. To develop this survey, LS coordinated with the National Cancer Institute, the Office of Behavioral and Social Sciences Research, the Centers for Disease Control and Prevention, the American

Corresponding author: Tyler G. Ketterl, MD, MS, Division of Pediatric Hematology/Oncology, Seattle Children's Hospital, 4800 Sand Point Way NE, Seattle, WA 98105; tyler.ketterl@seattlechildrens.org

[1]Fred Hutchinson Cancer Research Center, Seattle, Washington; [2]Division of Pediatric Hematology/Oncology, Seattle Children's Hospital, Seattle, Washington; [3]Jonsson Comprehensive Cancer Center, University of California Los Angeles, Los Angeles, California; [4]Abramson Cancer Center, University of Pennsylvania, Philadelphia, Pennsylvania; [5]Memorial Sloan-Kettering Cancer Center, New York, New York; [6]University of Colorado Cancer Center, Denver, Colorado; [7]Dana-Farber Cancer Institute, Boston, Massachusetts; [8]University of North Carolina Lineberger Comprehensive Cancer Center, Chapel Hill, North Carolina.

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Cancer Society, and the Agency for Healthcare Research and Quality to include elements of the Experiences with Cancer Survivorship supplement of the Medical Expenditures Panel Survey.[8] More information about the survey can be found on the LS Foundation's website (https://www.livestrong.org/what-we-do/our-research. Accessed January 16, 2019) and related publications on the population of the original LS survey.[9,10]

To address the gap in knowledge regarding the long-term and late effects of cancer specifically in the AYA population, we used the LS survey paired with treatment data abstracted from medical records at 7 national comprehensive cancer centers that were members of the LS Survivorship Centers of Excellence Network. The objective of the current study was to evaluate the impact of cancer-directed treatment on physical and mental impairment of work-related tasks, the need for employment changes, including paid or unpaid time off from work, and the financial toxicity for cancer survivors aged 18 to 39 and their families.

MATERIALS AND METHODS

Recruitment, Study Setting, and Participants

This multicenter, cross-sectional study of AYA cancer survivors was conducted through the LS Survivorship Centers of Excellence Network, which is composed of 7 geographically diverse sites, including the Fred Hutchinson Cancer Research Center/Seattle Cancer Care Alliance (coordinating center) in Seattle, the Abramson Cancer Center at the University of Pennsylvania in Philadelphia, the Dana Farber Cancer Institute in Boston, the Memorial Sloan-Kettering Cancer Center in New York City, the Jonsson Comprehensive Cancer Center at the University of California in Los Angeles, the University of Colorado Cancer Center in Denver, and the University of North Carolina in Chapel Hill. Human investigations were performed after approval by the Fred Hutchinson Cancer Research Center Institutional Review Board. Eligibility for study participation included the diagnosis of a malignancy between ages 18 and 39 years, survey completion within 1 to 5 years from diagnosis and ≥1 year after therapy completion. Participants were sent a recruitment letter and brochure; if a participant did not respond, then they were sent a second recruitment letter 7 to 10 days after the initial letter and were called up to 3 times with up to 3 voice mail contacts or voice messages attempted. Those who agreed to participate were then sent an e-mail link to complete the study registration online. Once they were registered to participate, enrolled participants were subsequently sent an e-mail with a link containing a unique URL for the study informed consent form and survey.

Assessment of Physical, Mental, and Financial Effects

Participants were asked to complete an online survey of patient reported outcomes (PROs). Questions from the PRO related to physical and mental impairment of work-related tasks, extended paid or unpaid time off from work, and cancer survivors or their families borrowing money or going into debt. Questions also were asked about the highest level of education achieved and insurance status. For the current investigation, we focused on responses to the following subset of questions: 1) Did you ever feel that your cancer, its treatment, or the lasting effects of that treatment interfered with your ability to perform any physical tasks required by your job? 2) Did you ever feel that your cancer, its treatment, or the lasting effects of that treatment interfered with your ability to perform any mental tasks required by your job? 3) At any time since your first cancer diagnosis, did you take extended paid time off from work, unpaid time off, or make a change in your hours, duties, or employment status? 4) Did you ever take extended paid time off work (vacation, sick time, and/or disability leave)? 5) Did you ever take unpaid time off from work? 6) How much did you or your family borrow or how much debt did you incur because of your cancer, its treatment, or the lasting effects of that treatment (response options varied from <$10,000 to $100,000 or more).

Treatment Exposures

Treatment data were abstracted from medical records using a standardized protocol and were entered into a centralized, secured, online Survivorship Informatics Management System, which permitted single data entry, largely from menus with pick-lists, to maintain consistently coded, high-quality data with minimal missing data for key variables. Treatment data included information regarding surgery, chemotherapy, and/or radiation.

Covariates

Potential confounders of the associations between cancer-directed treatment exposure and physical, mental, and financial outcomes were selected a priori based on a literature review and included sex, age at diagnosis (categorized as ages 18-24, 25-35, or >35 years), health insurance coverage, and education (categorized as high school or less, some college/associates degree, college graduate, or postgraduate work).

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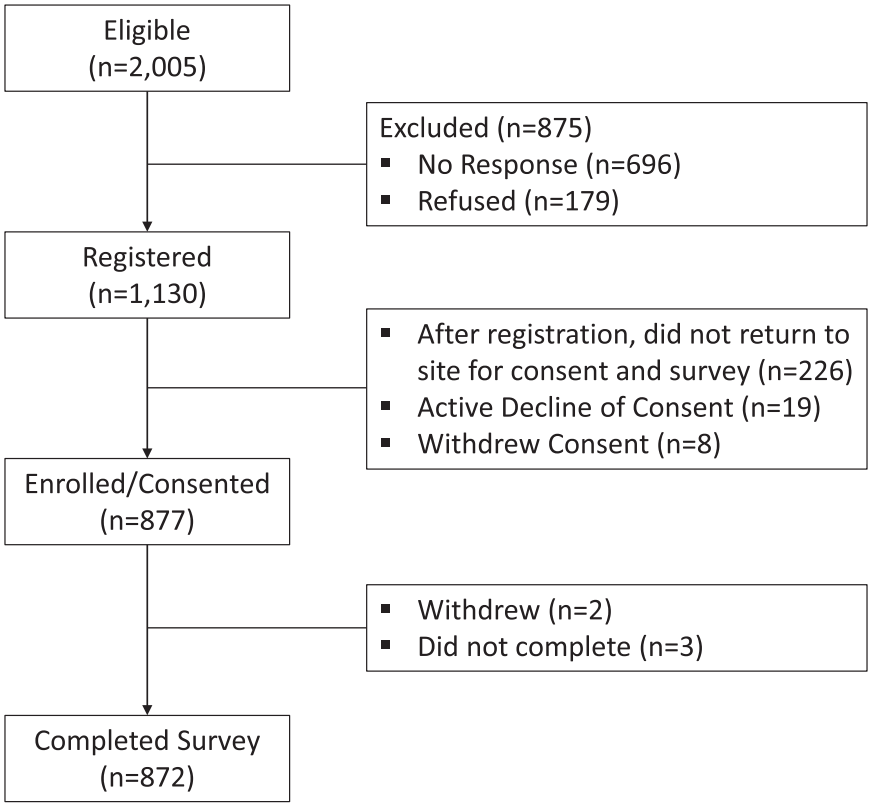

Figure 1. This is a Consolidated Standards of Reporting Trials (CONSORT) flow diagram of participation in the current study.

Statistical Analyses

Data were analyzed separately by tumor site, including: 1) breast, 2) thyroid, 3) leukemia and lymphoma, 4) and all other cancer sites. Within each tumor site, the PROs were analyzed using logistic regression; Wald-based 95% confidence intervals (CIs) were generated for all associations. Analyses of each cancer-directed treatment exposure (surgery vs no surgery, chemotherapy vs no chemotherapy, radiation vs no radiation) were controlled for age (by category), education (by category), insurance status, and sex, with the exception of breast cancer-specific analyses, all of which were among females. Breast cancer survivors were analyzed for combined exposure of chemotherapy and radiation versus chemotherapy alone versus radiation alone, and the analyses were controlled for age, insurance status, and education (by category). Leukemia and lymphoma survivors were evaluated for combined exposure of radiation and surgery versus radiation alone versus only surgery, and the analyses were controlled for age, sex, insurance status, and education (by category). Because of heterogeneity in the group with cancers other than breast cancer, leukemia, lymphoma, or thyroid cancer, analyses of multiple combinations of treatment factors were not performed. When >90% of participants had received surgery or chemotherapy, the analyses did not include that treatment in the controlled variables. Greater than 95% of breast cancer survivors underwent surgery as part of their treatment course; therefore, analyses among breast cancer survivors were controlled for treatment exposures other than surgery (chemotherapy or radiation). Similarly, greater than 90% of leukemia and lymphoma survivors were exposed to chemotherapy; therefore, analyses within this survivor population were adjusted for radiation or surgery, but not for chemotherapy. The analyses were limited to enrolled study participants who completed the PRO survey. All analyses were performed in RStudio (version 1.0.153; RStudio, Inc, Boston, MA).[11]

RESULTS

The study initially identified potential participants from 7 different study sites in tumor registries or patient schedules for follow-up. From a screening of medical records, 2005 eligible survivors were identified. Of those, 1309 survivors (65.3%) were successfully contacted, and

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1130 (56.4%) registered for the study, providing permission to receive the study survey link electronically or by mail if requested (N = 14). In response to the study e-mail with survey link (or to the mailed survey and consent), 877 survivors (67% of those contacted) consented, and 872 completed the survey (Fig. 1). Passive refusal and/or incorrect contact information was noted for 34.7% of potentially eligible survivors, and there were no tracing efforts of those individuals. Of the participants who completed the PRO survey, 635 (72.8%) were female, and 237 (27.2%) were male, with a median age at diagnosis of 32.3 and 29.8 years, respectively, and a mean of 3.53 and 3.40 years from diagnosis, respectively; and 80.6% of participants were white (Table 1). In addition to all but 1 breast cancer survivor being female (N = 240), the majority of endocrine cancer (predominantly thyroid) and skin cancer (predominantly melanoma) survivors were female (84.4% and 74.4%, respectively), consistent with proportions of those diagnosed.[12]

Overall, 736 participants (84.4%) were employed at some time between their cancer diagnosis and study participation. Among the participants who reported working, 517 (70.2%) reported a physical component required by their job, and 303 (58.6%) reported that their cancer, its treatment, or the lasting effects of treatment interfered with physical tasks required by their job. Of the participants who were working, 399 (54.2%) reported that their cancer, its treatment, or the lasting effects of treatment interfered with their ability to perform mental tasks required by their job. Among AYA survivors who were working and reported that physical tasks were associated with their jobs, those who had been exposed to chemotherapy were significantly more likely to report interference with those job-related physical tasks (odds ratio [OR], 1.97; 95% CI, 1.22-311; $P < .01$) compared with survivors who did not receive chemotherapy, adjusting for radiation and surgery exposure as well as age at diagnosis and sex (Table 2). Receipt of chemotherapy also was associated significantly with reported interference with mental tasks required by a job (OR, 3.22; 95% CI, 2.15-4.79; $P < .01$) and any time off from work (OR, 3.56; 95% CI, 2.31-5.47; $P < .01$). With the exception of an association between radiation and interference with physical tasks at work, the associations for radiation and surgery were not statistically significant (Table 2). Across diagnoses, 126 AYA survivors (14.4%) reported that their families borrowed ≥$10,000. Those who were exposed to chemotherapy were significantly more likely to report that they or their families borrowed ≥$10,000 (OR, 3.05; 95% CI, 1.53-6.09; $P < .01$) compared with survivors who were not exposed to chemotherapy within the same groups according to sex, age (categorical), education (categorical), insurance status, radiation exposure, and surgery exposure. In total, 13 AYA survivors (1.5%) reported that they or their family had filed for bankruptcy because of their cancer, its treatment, or the lasting effects of that treatment. Because of the infrequent rate of this outcome, we did not evaluate associations with treatment modalities.

TABLE 1. Characteristics of Young Adult Cancer Survivors by Sex

Variable	No. of Survivors (%)	
	Female	Male
Total no.	635	237
Age at diagnosis: Mean ± SD, y	32.3 ± 5.62	29.8 ± 6.09
Age at diagnosis, y		
18-24	84 (13.2)	55 (23.2)
25-34	306 (48.2)	122 (51.5)
35-39	245 (38.6)	60 (25.3)
Time from diagnosis: Mean ± SD, y	3.53 ± 1.49	3.40 ± 1.29
Ethnicity		
Not Hispanic or Latino	577 (90.9)	212 (89.5)
Hispanic or Latino	49 (7.7)	23 (9.7)
Declined/unknown	9 (1.4)	2 (0.8)
Race		
White	504 (79.4)	199 (84.0)
Asian/Pacific Islander	57 (9.0)	21 (8.9)
Black	43 (6.8)	6 (2.5)
Deferred/NA	31 (4.9)	11 (4.6)
Education		
≤High school	41 (6.5)	17 (7.2)
Some college/associates degree	126 (19.8)	59 (24.9)
College graduate	231 (36.4)	89 (37.6)
Postgraduate work	215 (33.9)	64 (27.7)
Insurance coverage	605 (95.3)	219 (92.4)
Cancer diagnosis		
Breast	240 (37.8)	1 (0.4)
Leukemia and lymphoma	83 (13.1)	80 (33.8)
Endocrine system[a]	108 (17.0)	20 (8.4)
Skin	60 (9.4)	21 (8.9)
Genital system, female[b]	49 (7.7)	0 (0.0)
Genital system, male[c]	0 (0.0)	46 (19.4)
Brain and other nervous system	24 (3.8)	17 (7.2)
Bones and soft tissue	17 (2.7)	19 (8.0)
Digestive system[d]	24 (3.8)	11 (4.6)
Oral cavity and pharynx	13 (2.0)	12 (5.1)
Urinary system[e]	8 (1.3)	6 (2.5)
Other	10 (1.6)	3 (1.3)

Abbreviations: NA, not available; SD, standard deviation.

[a]The endocrine system included the thyroid, adrenals, and other endocrine structures.

[b]The female genital system included the uterine cervix, uterine corpus, vagina, vulva, and other female structures.

[c]The male genital system included the prostate, testis, penis, and other male genital structures.

[d]The digestive system included the esophagus, small intestine, colon, rectum, anus, anal canal, anus/rectum, liver, intrahepatic biliary duct, hepatoblastoma, gallbladder, biliary system, pancreas, and other digestive organs.

[e]The urinary system included the urinary bladder, kidney, renal pelvis, ureter, Wilms tumor, and other urinary organs.

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TABLE 2. Association of Cancer-Directed Treatment Factors and Physical and Mental Impairment at Work With Changes in Employment Status Among Young Adult Cancer Survivors Who Worked at Any Time Between Diagnosis and Study Participation

	Chemotherapy[a]			Radiation[b]			Surgery[c]		
	No. Yes/No (% Yes)	OR	95% CI	No. Yes/No (% Yes)	OR	95% CI	No. Yes/No (% Yes)	OR	95% CI
Variable									
Interfered with physical tasks required by job									
No	96/119 (44.7)	1.00	Ref	80/135 (37.2)	1.00	Ref	172/43 (80.0)	1.00	Ref
Yes	187/115 (61.9)	1.97	1.22-3.11[d]	140/162 (46.4)	1.66	1.08-2.41[e]	240/62 (79.5)	1.57	0.90-2.73
Interfered with mental tasks required by job									
No	144/189 (43.2)	1.00	Ref	122/211 (36.6)	1.00	Ref	264/69 (79.3)	1.00	Ref
Yes	269/130 (67.4)	3.22	2.15-4.79[d]	182/217 (45.6)	1.39	0.99-1.94	321/78 (80.5)	1.46	0.90-2.35
Extended paid time off from work, unpaid time off, or a change in hours, duties, or employment status									
No	64/108 (37.2)	1.00	Ref	63/109 (36.6)	1.00	Ref	137/35 (79.7)	1.00	Ref
Yes	352/212 (62.4)	3.56	2.31-5.47[d]	243/321 (43.1)	1.07	0.73-1.58	452/112 (80.1)	1.21	0.71-2.08
Extended paid time off from work									
No	115/63 (64.6)	1.00	Ref	77/101 (43.3)	1.00	Ref	143/35 (80.3)	1.00	Ref
Yes	228/138 (62.3)	1.02	0.62-1.67	165/201 (45.1)	1.03	0.69-1.55	293/73 (80.1)	0.54	0.54-1.00[e]
Unpaid time off from work									
No	159/96 (62.4)	1.00	Ref	119/136 (46.7)	1.00	Ref	199/56 (78.0)	1.00	Ref
Yes	183/103 (64.0)	1.39	0.79-2.36	122/164 (42.7)	0.90	0.62-1.32	234/52 (81.8)	1.37	0.79-2.36

Abbreviations: CI, confidence interval; OR, odds ratio; Ref, reference category.
[a]The analysis was adjusted for diagnosis (categorical), age (categorical), sex, education (categorical), insurance, radiation status, and surgery status.
[b]The analysis was adjusted for diagnosis (categorical), age (categorical), sex, education (categorical), insurance, chemotherapy status, and surgery status.
[c]The analysis was adjusted for diagnosis (categorical), age (categorical), sex, education (categorical), insurance, chemotherapy status, and radiation status.
[d]$P < .01$.
[e]$P < .05$.

Breast Cancer

In total, 241 participants who completed the PRO survey had a diagnosis of breast cancer, of whom 209 (86.7%) were employed at some time between cancer diagnosis and study participation. Among survivors who were working, those who had been exposed to chemotherapy were significantly more likely to report interference with mental tasks required by their job (OR, 2.66; 95% CI, 1.32-5.34; $P < .01$) and taking unpaid time off from work (OR, 2.62; 95% CI, 1.09-6.30; $P < .05$) compared with those who had not been exposed to chemotherapy (Table 3). Also, those who had been exposed to radiation were significantly more likely to report interference with mental tasks required by their job (OR, 2.05; 95% CI, 1.11-3.78; $P < .05$) compared with those who had not been exposed to radiation. We did not observe an association with cancer-directed treatment exposure and the ability to perform physical tasks required by a job, nor was there an association between cancer-directed treatment exposure and AYA breast survivors or their families borrowing >$10,000. Among breast cancer survivors with the same age, education, and insurance status who were working, those who had received chemotherapy were more likely to take extended paid time off work compared with those who had received both chemotherapy and radiation (OR, 3.08; 95% CI, 1.18-8.03; $P = .02$). Participants who had received chemotherapy and radiation also were more likely to have had advanced disease.

Leukemia and Lymphoma

Of the 163 survey participants who had a diagnosis of leukemia or lymphoma, 130 (79.8%) were employed at some time between their cancer diagnosis and study participation. Nearly all of these participants had received chemotherapy (90%), as noted above. No further association was observed between cancer-directed treatment exposure to surgery or radiation and interference with physical or mental tasks required by their job among working leukemia and lymphoma survivors (Table 4). In this group, there was no association between cancer-directed radiation or surgery exposures and paid time or unpaid time off from work, nor was there an association between cancer-directed treatment exposure

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TABLE 3. Association of Cancer-Directed Treatment Factors and Physical and Mental Impairment at Work With Changes in Employment Status Among Young Adult Breast Cancer Survivors Who Worked at Any Time Between Diagnosis and Study Participation

	Chemotherapy[a]			Radiation[b]		
Variable	No. Yes/No (% Yes)	OR	95% CI	No. Yes/No (% Yes)	OR	95% CI
Interfered with physical tasks required by job						
No	42/17 (71.2)	1.00	Ref	33/26 (55.9)	1.00	Ref
Yes	66/13 (83.5)	2.03	0.82-5.01	47/32 (59.5)	1.11	0.53-2.33
Interfered with mental tasks required by job						
No	57/33 (63.3)	1.00	Ref	41/49 (45.6)	1.00	Ref
Yes	100/18 (84.7)	2.66	1.32-5.34[c]	77/41 (65.3)	2.05	1.11-3.78[d]
Extended paid time off from work, unpaid time off, or a change in hours, duties, or employment status						
No	29/13 (69.0)	1.00	Ref	27/15 (64.3)	1.00	Ref
Yes	129/38 (77.2)	1.76	0.78-3.99	91/76 (54.5)	0.62	0.29-1.33
Extended paid time off from work						
No	38/8 (82.6)	1.00	Ref	36/10 (78.3)	1.00	Ref
Yes	88/28 (75.9)	1.10	0.42-2.91	55/61 (47.4)	0.25	0.10-0.62[c]
Unpaid time off from work						
No	60/25 (70.6)	1.00	Ref	44/41 (51.8)	1.00	Ref
Yes	65/11 (85.5)	2.62	1.09-6.30[d]	46/30 (60.5)	1.14	0.57-2.28

Abbreviations: CI, confidence interval; OR, odds ratio; Ref, reference category.
[a]The analysis was adjusted for age (categorical), education (categorical), insurance, and radiation status.
[b]The analysis was adjusted for age (categorical), education (categorical), insurance, and chemotherapy status.
[c]$P < .01$.
[d]$P < .05$.

TABLE 4. Association of Cancer-Directed Treatment Exposures to Radiation or Surgery and Physical and Mental Impairment at Work With Changes in Employment Status Among Young Adult Survivors of Leukemia or Lymphoma Who Worked at Any Time Between Diagnosis and Study Participation[a]

	Radiation[b]			Surgery[c]		
Variable	No. Yes/No (% Yes)	OR	95% CI	No. Yes/No (% Yes)	OR	95% CI
Interfered with physical tasks required by job						
No	13/13 (50.0)	1.00	Ref	6/20 (23.1)	1.00	Ref
Yes	27/29 (48.2)	1.01	0.34-2.94	20/36 (35.7)	1.63	0.49-5.44
Interfered with mental tasks required by job						
No	25/26 (49.0)	1.00	Ref	15/36 (29.4)	1.00	Ref
Yes	29/49 (37.2)	0.68	0.32-1.45	25/53 (32.1)	1.11	0.47-2.62
Extended paid time off from work, unpaid time off, or a change in hours, duties, or employment status						
No	12/12 (50.0)	1.00	Ref	7/17 (29.2)	1.00	Ref
Yes	43/63 (40.6)	0.61	0.23-1.58	34/72 (32.1)	0.77	0.25-2.39
Extended paid time off from work						
No	13/31 (29.5)	1.00	Ref	18/26 (40.9)	1.00	Ref
Yes	29/30 (49.2)	2.50	0.98-6.39	15/44 (25.4)	0.42	0.15-1.20
Unpaid time off from work						
No	25/28 (47.2)	1.00	Ref	14/39 (26.4)	1.00	Ref
Yes	17/33 (34.0)	0.54	0.22-1.31	19/31 (38.0)	1.54	0.59-4.02

Abbreviations: CI, confidence interval; OR, odds ratio; Ref, reference category.
[a]All patients received chemotherapy.
[b]The analysis was adjusted for age (categorical), sex, education (categorical), insurance, and surgery status.
[c]The analysis was adjusted for age (categorical), sex, education (categorical), insurance, and radiation status.

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TABLE 5. Association of Cancer-Directed Radiation and Physical and Mental Impairment at Work With Changes in Employment Status Among Young Adult Survivors of Thyroid Cancer Who Worked at Any Time Between Diagnosis and Study Participation[a]

	Radiation[b]		
Variable	No. Yes/No (% Yes)	OR	95% CI
Interfered with physical tasks required by job			
No	20/21 (48.8)	1.00	Ref
Yes	21/12 (63.6)	2.16	0.70-6.62
Interfered with mental tasks required by job			
No	28/27 (50.9)	1.00	Ref
Yes	29/20 (59.2)	1.69	0.72-3.96
Extended paid time off from work, unpaid time off, or a change in hours, duties, or employment status			
No	14/15 (48.3)	1.00	Ref
Yes	43/32 (57.3)	1.51	0.59-3.83
Extended paid time off from work			
No	8/13 (38.1)	1.00	Ref
Yes	35/15 (70.0)	5.05	1.36-18.73[c]
Unpaid time off from work			
No	22/15 (59.5)	1.00	Ref
Yes	21/13 (61.8)	1.11	0.31-3.95

Abbreviations: CI, confidence interval; OR, odds ratio; Ref, reference category.
[a]Chemotherapy is not standard for thyroid cancer treatment, and nearly all patients underwent surgery.
[b]The analysis was adjusted for age (categorical), sex, education (categorical), and insurance.
[c]$P < .05$.

and AYA leukemia and lymphoma survivors or their families borrowing >$10,000. Analyses were performed for joint associations between treatment exposures, including surgery and radiation, in the leukemia and lymphoma group, and no statistically significant associations were identified.

Thyroid Cancer

All but 2 endocrine cancers were diagnosed as thyroid cancer; therefore, the analyses included only 126 thyroid cancer survivors. Of those participants, 104 (82.5%) were employed at some time between their cancer diagnosis and study participation. No association was identified between cancer-directed treatment exposures and interference with physical or mental tasks required by their job among the thyroid cancer survivors (Table 5). Among thyroid cancer survivors who were working, those who had been exposed to radiation were significantly more likely to report taking extended paid time off from work (OR, 5.05; 95% CI, 1.36-18.73; $P < .05$) than those who had not been exposed to radiation. There was no association between cancer-directed treatment exposures and interference with paid or unpaid time off from work, nor was there an association between cancer-directed treatment exposure and AYA thyroid cancer survivors or their families borrowing >$10,000.

Other Cancers

The remaining 328 participants who completed the PRO survey had diagnoses other than breast cancer, leukemia, lymphoma or thyroid cancer, including sarcoma, germ cell tumors, skin cancer, and brain tumors. Of these 281 survivors (85.7%) were employed at some time between their cancer diagnosis cancer and study participation. Among survivors who were working, those who had been exposed to chemotherapy were significantly more likely to report interference with mental tasks required by their job (OR, 3.67; 95% CI, 2.16-6.24; $P < .01$) and reported taking extended paid time off from work, unpaid time off, or changes in hours, duties, or employment status (OR, 5.52; 95% CI, 2.78-10.94; $P < .01$) compared with those who had not been exposed to chemotherapy (Table 6). Survivors who had been exposed to radiation were significantly more likely to report interference with physical tasks required by their job (OR, 3.11; 95% CI, 1.41-6.88; $P < .01$) and taking extended paid time off from work, unpaid time off, or changes in their hours, duties, or employment status (OR, 2.31; 95% CI, 1.02-5.22; $P < .05$). Furthermore, among survivors who were working, those who underwent surgery were significantly more likely to report interference with mental tasks required by their job (OR, 2.01; 95% CI, 1.03-3.92; $P < .05$) compared with those who did not undergo surgery within the same groups according to sex, age (categorical), chemotherapy, and radiation exposure. Analyses were performed for joint associations between treatment exposure to surgery and radiation, and no statistically significant associations were identified. Finally, AYA survivors who had been exposed to chemotherapy were significantly more likely to report that they or their families borrowed >$10,000 (OR, 3.42; 95% CI, 1.50-7.83; $P < .01$) compared with survivors who had not been exposed to chemotherapy within the same groups according to sex, age (categorical), radiation exposure, and surgery exposure.

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TABLE 6. Association of Cancer-Directed Treatment Factors and Physical and Mental Impairment at Work With Changes in Employment Status Among Young Adult Cancer Survivors (Other Than Survivors of Breast Cancer, Leukemia, Lymphoma, and Thyroid Cancer) Who Worked at Any Time Between Diagnosis and Study Participation

	Chemotherapy[a]			Radiation[b]			Surgery[c]		
Variable	No. Yes/No (% Yes)	OR	95% CI	No. Yes/No (% Yes)	OR	95% CI	No. Yes/No (% Yes)	OR	95% CI
Interfered with physical tasks required by job									
No	31/58 (34.8)	1.00	Ref	14/75 (15.7)	1.00	Ref	68/21 (76.4)	1.00	Ref
Yes	68/66 (50.7)	1.70	0.89-3.23	45/89 (33.6)	3.11	1.41-6.88[d]	113/21 (84.3)	2.28	1.09-4.76[e]
Interfered with mental tasks required by job									
No	40/97 (29.2)	1.00	Ref	28/109 (20.4)	1.00	Ref	107/30 (78.1)	1.00	Ref
Yes	95/59 (61.7)	3.67	2.16-6.24[d]	47/107 (30.5)	1.56	0.84-2.92	134/20 (87.0)	2.06	1.05-4.02[e]
Extended paid time off from work, unpaid time off, or a change in hours, duties, or employment status									
No	14/63 (18.2)	1.00	Ref	10/67 (13.0)	1.00	Ref	60/17 (77.9)	1.00	Ref
Yes	122/94 (56.5)	5.52	2.78-10.94[d]	66/150 (30.6)	2.31	1.02-5.22[e]	183/33 (84.7)	1.70	0.85-3.40
Extended paid time off from work									
No	36/31 (53.7)	1.00	Ref	20/47 (29.9)	1.00	Ref	59/8 (88.1)	1.00	Ref
Yes	83/58 (58.9)	0.93	0.47-1.82	46/95 (32.6)	1.16	0.54-2.47	118/23 (83.7)	0.62	0.23-1.71
Unpaid time off from work									
No	47/33 (58.8)	1.00	Ref	28/52 (35.0)	1.00	Ref	66/14 (82.5)	1.00	Ref
Yes	72/54 (57.1)	1.19	0.63-2.28	38/88 (30.2)	0.86	0.43-1.70	109/17 (86.5)	1.47	0.62-3.51

Abbreviations: CI, confidence interval; OR, odds ratio; Ref, reference category.
[a]The analysis was adjusted for age (categorical), sex, education (categorical), insurance, radiation status, and surgery status.
[b]The analysis was adjusted for age (categorical), sex, education (categorical), insurance, chemotherapy status, and surgery status.
[c]The analysis was adjusted for age (categorical), sex, education (categorical), insurance, chemotherapy status, and radiation status.
[d]$P < .01$.
[e]$P < .05$.

DISCUSSION

In this cross-sectional study of AYA cancer survivors, among the entire study population, radiation was associated with increased odds of reporting physical impairments of work tasks, and chemotherapy was associated with an increased likelihood of reporting both physical and mental impairment of work tasks as well as an increased likelihood of taking any time off work. In subgroup analyses by diagnosis, we observed that chemotherapy was associated with increased odds of mental impairment of work tasks in AYA breast cancer and "other" cancer survivors, increased odds of taking paid time off from work in breast cancer survivors, increased odds of taking any time off work in AYA survivors of "other" cancers, and increased odds that AYA survivors of "other" cancer survivors or their families would borrow >$10,000. Exposure to radiation therapy was associated with increased odds of mental impairment of work tasks among AYA breast cancer survivors, increased odds of taking an extended paid time off work among AYA thyroid cancer survivors, and increased odds of physical impairment and taking any time off work among AYA survivors of "other" cancers. Exposure to a surgical procedure was associated with increased odds of physical and mental impairment of work tasks among AYA survivors of "other" cancers, including survivors of sarcoma and central nervous system cancers. Finally, among breast cancer survivors who were working, those who had received chemotherapy were more likely to take extended time off work compared with those who had received both chemotherapy and radiation. This association may be explained in part by our inability to determine which participants stopped working versus those who took extended paid time off. Participants who stopped working were not captured by the survey questions. This may have resulted in the chemotherapy-only group reporting having taken more extended time off, whereas those who had received combined chemotherapy and radiation actually may have stopped working entirely, but this would not have been captured by the survey. Supporting this explanation, those who had received combined chemotherapy and radiation also were more likely to have had advanced disease.

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Although there has been a considerable amount of research specifically addressing cancer and work outcomes, most has focused on the likelihood and timeliness of returning to work after cancer.[13,14] The Adolescent and Young Adult Health Outcomes and Patient Experience (AYA HOPE) Study evaluated the impact of cancer on work and education in a cohort of 463 patients aged 15 to 39 years who were recently diagnosed with cancer from participating Surveillance, Epidemiology, and End Results cancer registries. Those authors reported that, although most AYA patients with cancer returned to work after cancer, >50% continued to report some problems with work/studies on return.[15] The authors also demonstrated that, among full-time workers/students before diagnosis, those who were diagnosed with acute lymphocytic leukemia and non-Hodgkin lymphoma were less likely than those who had other cancer diagnoses to be working/in school at follow-up.[15] These results support the finding that leukemia and lymphoma survivors who received radiation were more likely to take extended time off work because of their cancer. Studies using data from the Childhood Cancer Survivor Study that have evaluated the issue of returning to work among adult survivors of childhood cancer indicate that survivors have higher levels of unemployment because of health or are between jobs more often than their siblings. In addition, cranial radiotherapy doses ≥25 grays were associated with higher odds of unemployment.[16] Prior data from patients with breast cancer also have indicated that the type of treatment is the principal determinant in the time taken to return to work.[17] A study of Korean breast cancer survivors demonstrated that fatigue and exhaustion were the most frequent difficulties encountered by survivors during occupational work.[18] We build on these findings, specifically for young adults, by identifying the treatment factors that affect physical and mental impairment of work tasks as well the need to take time off work. Our study fills a gap in the literature as it relates to AYA cancer survivors, the impact of cancer-directed therapies on employment and financial outcomes in these AYA cancer survivors, and the physical and mental impairment they may face.

AYA cancer survivors are a heterogeneous group; therefore, the treatments they receive and the morbidities from those treatments also are likely to be heterogeneous. It is possible that the associations we observe in the different cancer categories are related to advanced disease, which requires an intensification of therapy. For example, in the group that had diagnoses other than breast cancer, leukemia, lymphoma, or thyroid cancer, the association between radiation and physical limitations at work may have been caused by radiation exposure related to the treatment of brain tumors.

The results of this study should be interpreted in the context of study limitations. Because this was a 1-time, cross-sectional study, our results did not capture the trajectories of impairments in the cancer survivor population. We also lacked information regarding participants' insurance status, which may have a financial impact on survivors. Furthermore, there were limitations related to the lower study enrollment rate. Although this presents the possibility of selection bias and could impact generalizability, this response rate is similar to that observed in other studies of AYA cancer survivors, such as the AYA HOPE Study.[19] Limited study resources prevented any follow-up tracing efforts of participants who lacked a correct address or phone number. In addition, we acknowledge that there is potential for selection bias in this study, because the survivors who did not complete the survey may have been systematically different from those who completed the survey with respect to outcomes. It is also important to acknowledge that self-reported limitations, either physical or mental, are not equivalent to demonstrated measures of performance and may affect the results of self-reported, chemotherapy-related, cognitive impairment compared with formal neurocognitive testing. For analyses of exposure to chemotherapy, power and data were not sufficient to distinguish participants according to the chemotherapy agent or regimen received, and it is possible that some chemotherapy regimens may have more long-term consequences than others. Finally, it is also possible that treatment exposures could be a proxy for disease severity, and advanced disease may actually have a direct impact on mental and physical impairment at work. Patients who had more advanced disease may have received more intense therapy, and it is possible that symptoms of the disease itself, rather than direct effects of the treatment, may partially explain the observed associations.

In summary, to our knowledge, our study is the first describing the associations of cancer-directed therapy with employment and financial implications in a multicenter, large cohort of AYA cancers survivors. This work is important for understanding the challenges unique to AYA cancer survivors as they relate to employment during and after cancer therapy. The current results suggest that further longitudinal studies and interventions are needed to describe and support AYA cancer survivors in the work force during and after cancer-directed therapy.

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CONFLICT OF INTEREST DISCLOSURES

The authors have no disclosures.

AUTHOR CONTRIBUTIONS

Tyler G. Ketterl: Data curation, formal analysis, investigation, methodology, writing–original draft, and writing–review and editing. **Karen L. Syrjala:** Conceptualization, funding acquisition, data curation, and writing–review and editing. **Jacqueline Casillas:** Data curation and writing–review and editing. **Linda A. Jacobs:** Data curation and writing–review and editing. **Steven C. Palmer:** Data curation and writing–review and editing. **Mary S. McCabe:** Data curation and writing–review and editing. **Patricia A. Ganz:** Data curation and writing–review and editing. **Linda Overholser:** Data curation and writing–review and editing. **Ann Partridge:** Data curation and writing–review and editing. **Emily Jo Rajotte:** Project administration and writing–review and editing. **Betsy Risendal:** Data curation and writing–review and editing. **Abby R. Rosenberg:** Data curation and writing–review and editing. **Donald Rosenstein:** Data curation and writing–review and editing. **Kevin Scott Baker:** Conceptualization, funding acquisition, methodology, supervision, data curation, and writing–review and editing.

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